Description

Image





Caption

1. Zinc bucket. © Jon Pallbo, Pallbo at en.wikipedia - Public domain 2. Close-up of the Zinc bucket. © Jon Pallbo, Pallbo at en.wikipedia - Public domain 3. Guard rail along the road. © Pudding4brains at en.wikipedia - Public domain

The material

Zinc is a bluish-white metal with a low melting point (420 C). The slang in French for a bar or pub is "le zinc"; bar counters in France used to be clad in zinc - many still are - to protect them from the ravages of wine and beer. Bar surfaces have complex shapes - a flat top, curved profiles, rounded or profiled edges. These two sentences say much about zinc: it is ductile; it is hygienic; it survives exposure to acids (wine), to alkalis (cleaning fluids), and to misuse (upset customers). These remain among the reasons it is still used today. Another is the "castability" of zinc alloys - their low melting point and fluidity gives them a leading place in die-casting.

Compositional summary

99.5% Zn

General properties

Density	7.13e3	-	7.15e3	kg/m^3
Price	* 2.07	-	2.2	GBP/kg
Date first used	1746			

Mechanical properties

Young's modulus	* 90	-	110	GPa
Shear modulus	* 35	-	45	GPa
Bulk modulus	* 55	-	75	GPa
Poisson's ratio	0.245	-	0.255	
Yield strength (elastic limit)	* 90	-	150	MPa
Tensile strength	* 105	-	165	MPa
Compressive strength	* 90	-	150	MPa

Elongation	* 40	-	80	% strain
Hardness - Vickers	* 25	-	45	HV
Fatigue strength at 10^7 cycles	* 15	-	45	MPa
Fracture toughness	* 55	-	115	MPa.m^0.5
Mechanical loss coefficient (tan delta)	* 0.0017	-	0.0063	

Thermal properties

Melting point	418	-	420	°C
Maximum service temperature	* 79.9	-	99.9	°C
Minimum service temperature	* -20.2	-	-0.15	°C
Thermal conductor or insulator?	Good conductor			
Thermal conductivity	* 110	-	125	W/m.°C
Specific heat capacity	385	-	395	J/kg.°C
Thermal expansion coefficient	25	-	31	µstrain/°C

Electrical properties

Electrical conductor or insulator?	Good conductor			
Electrical resistivity	5.4	-	6	µohm.cm

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	Yes

Processability

Castability	5			
Formability	2	-	3	
Machinability	5			
Weldability	3	-	4	
Solder/brazability	4	-	5	

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Acceptable
Soils, acidic (peat)	Acceptable
Soils, alkaline (clay)	Excellent
Wine	Acceptable

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Limited use
Citric acid (10%)	Limited use
Hydrochloric acid (10%)	Unacceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Limited use
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Acceptable
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Excellent
Formaldehyde (40%)	Acceptable

Glycerol	Acceptable
Methyl alcohol (methanol)	Limited use

Durability: halogens and gases

Chlorine gas (dry)	Acceptable
Fluorine (gas)	Acceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Limited use

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Acceptable
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.15e7	tonne/yr
Reserves, principal component	2.3e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 3.49	-	3.84	kg/kg
Embodied energy (virgin grade production)	* 43.2	-	47.6	MJ/kg
Water usage	* 327	-	361	l/kg

Material processing: energy

Metal casting, embodied energy	* 34.5	-	38.6	MJ/kg
Metal extrusion, hot, embodied energy	* 5.78	-	9.95	MJ/kg
Metal rolling and forging, embodied energy	* 3.35	-	5.49	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg

Metal powder forming, embodied energy	* 17.8	-	20.2	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 3.7e3	-	4.71e3	MJ/kg
Machining, coarse, embodied energy	* 15.3	-	17.5	MJ/kg
Machining, fine, embodied energy	* 16.8	-	20.2	MJ/kg
Machining, grinding, embodied energy	* 18.3	-	22.9	MJ/kg
Machining, laser, CO2, embodied energy	* 0.441	-	0.561	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 2.35	-	2.64	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 0.378	-	0.652	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.219	-	0.36	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 1.41	-	1.6	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 305	-	387	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1	-	1.14	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.1	-	1.32	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.2	-	1.5	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0363	-	0.0461	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 1.12	-	1.23	kg/kg
Embodied energy, recycling	* 17.3	-	19	MJ/kg
Recycle fraction in current supply	22			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Zinc vapor is toxic - if you inhale it you get the "spelter-shakes" - but adequate protection is now universal. In all other ways zinc is a star: it is non-toxic, has low energy content, and - in bulk - can be recycled (not, of course, as plating).

Supporting information

Design guidelines

Zinc is used as cladding and in galvanizing steel to improve corrosion resistance. Wrought zinc is available as strip, sheet, foil, plate, rod, wire and blanks for forging or extrusion. It is relatively soft (a strength of 60 - 120 MPa) but, because of its hexagonal structure, bends in rolled zinc sheet should be at right angles to the grain or rolling direction and should have a radius no less than the sheet thickness. Wrought zinc alloys are easily soldered and spot-welded. It can be polished, textured, plated or painted. When exposed to air, zinc develops a surface film of carbonate that is very protective; it is frequently left uncoated.

Technical notes

Wrought zinc is made by hot-rolling cast sheets, by extrusion or by drawing. Zinc foil is made by electroplating zinc on an aluminum drum and then stripping it off.

Typical uses

Galvanizing of steel and other protective plating, cladding, flashing and guttering of buildings, flashlight reflectors, radio shielding, gaskets, photo-engraving plates, kitchen counter-tops, electrodes for zinc-carbon batteries.

Links

ProcessUniverse

Producers

Reference